Resilient Adaptive Stabilization for Nonlinear CPSs Dealing With Deception Attacks

Xinjun Wang, Ye Cao, Ben Niu, Huanqing Wang, Guangdeng Zong, *Senior Member, IEEE*, and Xudong Zhao

***Abstract*—It is still an open problem to output feedback stabilization with input quantizations and deception attacks for nonlinear cyber-physical systems subject to mismatched parametric uncertainties via backstepping design. This is because: first, the system contains mismatched uncertainties; second the output is corrupted by an additional attack signal, making the conventional recursive control design strategy using traditional error surfaces unsuitable; and third, since only the compromised output signal is available, the impacts of nonlinear items related to unknown attack weights become difficult to be explicitly addressed in the existing available output feedback control results. In this article, we present a solution to circumvent these obstacles by constructing a set of new K-filters using compromised output only, which, together with a coordinate transformation involving the attacked output and filter variables, allow a new form of state-observer-based resilient adaptive control scheme to be developed. Meanwhile, a novel compensation term is incorporated into the real controller to eliminate the effect of quantization on the system stability. It is shown that, by establishing Lyapunov functions appropriately, the original output of the system converges to zero, and all the signals in the closed-loop system are globally uniformly bounded. A practice example is provided to confirm the effectiveness of the proposed strategy.**

***Index Terms*—Adaptive backstepping control, deception attacks, K-filters, nonlinear cyber-physical systems (CPSs).**

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I. Introduction

In RECENT years, with the openness of information interactions and the complexity of physical dynamics, the cybersecurity problem has captured the great noticed of the control field [1], [2], [3]. The cyber-physical systems (CPSs) are vulnerable to malicious attackers on the network due to the highly openness of data transmission between controllers and actuators. In general, there are three typical attacks against CPSs, i.e., denial-of-service (DOS) attacks [4], replay attacks [5], and deception attacks [6], [7]. The DOS attacks can make computers or networks unable to provide normal services. Replay attacks repeat or delay the communication data to disrupt system stability. The purpose of deception attacks is to influence the data integrity by the injection of false information into CPSs; thus, the information transmitted by networks is maliciously attacked by additive signals. The possible locations of the three types of attacks mentioned earlier during the control process are shown in Fig. 1. Among them, the deception attacks are more dangerous and complicated compared with the other two attack types in an insecure network environment, which might lead to instability by modifying the transmitted data. Therefore, developing the resilient control strategy subject to unknown deception attacks is a critical issue. Until now, some meaningful resilient adaptive control results about the deception attacks have been devoted for linear systems [8], [9]. However, these adaptive resilient control approaches [8], [9] cannot be applied to the more general case of high-order nonlinear CPSs with mismatched parametric uncertainties. It is well known that the backstepping technique is an effective means to handle control problems of nonlinear systems with highly unknown and unmatched nonlinearities [10], [11]. The important point is that for the security problems of nonlinear systems suffering from malicious attacks, since the state signals are damaged by attacks, only the wrong states can be obtained and employed directly to design adaptive control schemes. Thus, the research on resilient adaptive backstepping control for the nonlinear systems with deception attacks is a challenging task since the traditional error transformation methods [10], [11] are not suitable for the controller design and only the compromised

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states are available. More recently, some adaptive backstepping-based resilient control results against deception attacks have been proposed for uncertain nonlinear systems [6], [12], [13], [14], [15]. It is noteworthy that the aforementioned works [6], [12], [13], [14], [15] are all assumed that the full states are available, which limits their applications. Focusing on the issue of output feedback control under deception attacks is undoubtedly meaningful although challenging. In [16] and [17], a novel state observer was constructed based on the compromised output, and a resilient adaptive controller was designed for nonlinear CPSs under deception attacks. However, the attack signal in [16] and [17] was simply reduced to an unknown bounded constant by using the inequality method, and the impact of attack term still exists. In other words, how to design a suitable state observer under the sensor and actuator deception attacks is still a challenging work. In addition, when communication channels are used for signal transmission in practical applications, it is very necessary and important to consider the signal quantization, and many fundamental results on quantized control have been addressed in [18], [19], and [20]. However, considering the cybersecurity problems encountered by the system in the process of signal transmission under deception attacks, the output measured by the sensor is corrupted, that is, the false output signal is applied to design controllers recursively; thus, the quantized control input is based on a false state variable, rendering those existing Lyapunov-stability-based resilient recursive design methods inapplicable.

Motivated by the aforementioned limitations, this study aims to solve the problem of resilient adaptive quantized control for nonlinear CPSs under the sensor and actuator deception attacks. Compared with those existing relevant results, the main contributions of our work are as follows.

1) Compared with the existing results [8], [9] that focus on linear CPSs with unknown deception attacks, here, a more general class of nonlinear CPSs with mismatched parametric uncertainties is considered. In addition, most of the existing resilient control-related results [6], [12], [13], [14], [15] are contingent upon the availability of complete state information. The presented control strategy in this article is extended to more general CPSs based on output feedback, rendering the control design and stability analysis complicated.
2) In contrast to the current observer based resilient control results [16], [17], where the impact of time-varying attack signal is neglected and the attack term is only treated as an unknown bounded constant, the malicious impact of unknown deception attacks is compensated with the aid of new K-filters, with which asymptotic stability can be achieved under deception attacks in this article.
3) Due to the unknown deception attacks appearing in the output of the system, the traditional error transformation of the backstepping design procedure in [10] and [11] is not available. Here, a new coordinate transformation involving error variables with the compromised output of the filters is constructed, which promotes a recursive control design strategy.

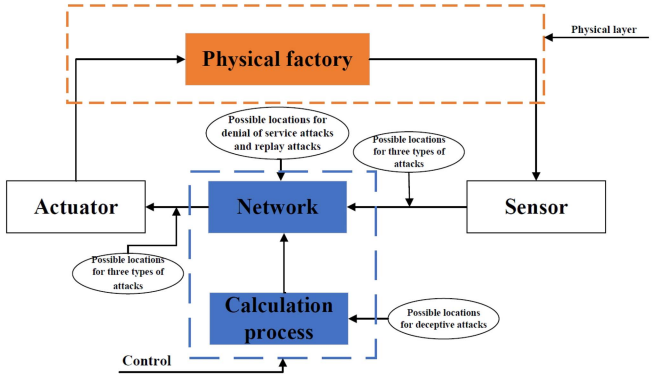


Fig. 1. Three possible locations for attacks during the control process.

4) Compared with the existing results [6], [12], [13], [14], [15], this article considers not only the sensor and actuator deception attacks but also input quantization, by designing an effective compensation term in the controller to eliminate the nonlinearities related to the quantizer, rendering the resultant resilient controller more practical in engineering application.

II. Problem Formulation and Preliminaries

In this section, only the output information after attack can be used for controller design, and the CPSs are modeled by following n-order uncertain nonlinear systems with unknown deception attacks and input quantizations:

$$\begin{aligned}\dot{x} &= Ax + \phi(y)\theta + \psi(y) + \begin{bmatrix} 0_{(r-1)\times 1} \\ b \end{bmatrix}[q(u) + \rho_a(t, x_1)] \\ \check{x}_1 &= x_1 + \omega_s(t, x_1) \\ y &= x_1 \end{aligned} \tag{1}$$

where $A = \begin{bmatrix} 0 & 1 & & & \\ 0 & 0 & 1 & & \\ \vdots & \vdots & & \ddots & \\ 0 & 0 & \cdots & 0 & 1 \\ 0 & 0 & \cdots & 0 & 0 \end{bmatrix} \in R^{n\times n}$, $x = [x_1, \ldots, x_n]^T \in R^n$ stand for the system states, and $y \in R$ and $u \in R$ are the system output and input, respectively. $\phi(y) \in R^{n\times p}$ and $\psi(y) \in R^n$ are the known smooth nonlinear functions. $\theta \in R^p$ are the vectors of unknown parameters, $q(u)$ denotes the output of the quantizer, $b = [b_m, \ldots, b_0]^T \in R^{m+1}$ are unknown constant vectors, and the sign of b_m, i.e., $\text{sign}(b_m)$, is also unknown. ρ_a is an actuator deception attack, $\omega_s(t, x_1)$ is the sensor deception attack, and $\check{x}_1$ is the available system state after attack.

Fig. 2 shows that the output of the system and the quantizer are attacked by unknown deception attacks. By monitoring the transmission on the state and controller information, attackers maliciously inject false data into the networks, thus destroying the original output and control signals of the system. In addition, although $\phi(\cdot) \in R^{n\times p}$ and $\psi(\cdot) \in R^n$ are known for

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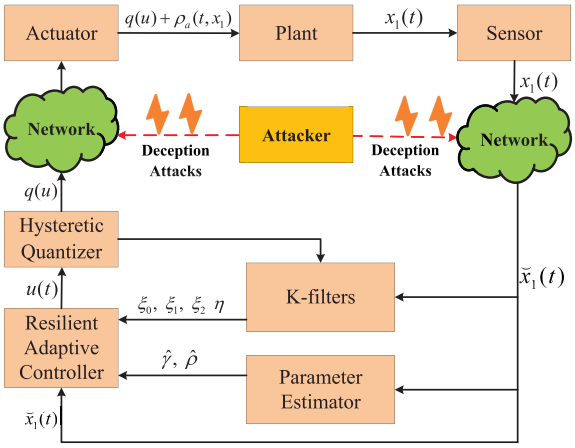


Fig. 2. Control block diagram for nonlinear systems with deception attacks.

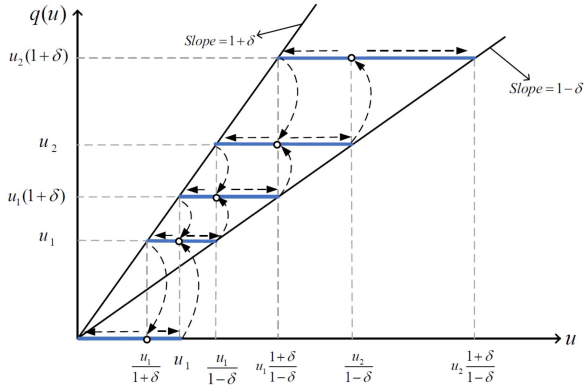


Fig. 3. Map of the hysteretic quantizer $q(u)$ for $u>0$.

the controller design, $\phi(y)$ and $\psi(y)$ become unknown since y is not available in the presence of cyberattacks, which brings additional difficulty in the process of observer and backstepping design, and further discussion will be held later.

The objective of this article is to determine the appropriate state estimator and resilient adaptive control scheme for uncertain nonlinear system (1) by utilizing the false output variable $\breve{x}_1$ compromised by attacks such that the output y converged to zero, and all the signals in the closed-loop system are bounded.

To achieve the aforementioned control purposes, the following assumptions are needed.

Assumption 1: $B(s)$ is a Hurwitz polynomial, i.e., all zeros of the polynomial $b_m s^m+\cdots+b_1 s+b_0$ are stable, and the relative degree $r(=n-m)$ is known.

Assumption 2 (see [6]): The terms $\omega_s(t,x_1)$ and $\rho_a(t,x_1)$ caused by deception attacks are defined as $\omega_s(t,x_1)=\zeta(t)x_1(t)$ and $\rho_a(t,x_1)=w_a(t)\Delta(x_1)$ with unknown time-varying signals $\zeta(t)(\zeta(t)\neq -1)$, $w_a(t)$, respectively. $\Delta(x_1)$ is a known smooth nonlinear function, and there exist unknown constants $\bar{\zeta}$, $\bar{\zeta}_d$, and $\bar{w}_a$ such that $|\zeta(t)|\leq\bar{\zeta}$, $|\dot{\zeta}(t)|\leq\bar{\zeta}_d$, and $|w_a(t)|\leq\bar{w}_a$.

Define $\lambda=(1+\zeta)^{-1}$; by recalling Assumption 2, we have $\lambda_m\leq|\lambda|\leq\lambda_M$ and $|\dot{\lambda}|\leq\bar{\dot{\lambda}}$, where λ_m,λ_M, and $\bar{\dot{\lambda}}$ are unknown positive constants. According to the definition of λ, the formula $x_1=\lambda\breve{x}_1$ holds.

Remark 1: In comparison with the existing adaptive secure control results in [6] and [15], where the sign $1+\zeta(t)>0$ for the attack weight is needed, here, the constraint condition on $\zeta(t)$ is relaxed in Assumption 2. Besides, from the description of system (1) and Assumption 2, we can get that $\breve{x}_1=(1+\zeta(t))x_1$. Thus, the condition $1+\zeta(t)\neq 0$ is to avoid the singularity problem, causing instability of the system.

Remark 2: In [21] and [22], the observer-based fault-tolerant control method is developed for a class of nonlinear strict-feedback systems with a sensor fault, where the sensor fault ζ is assumed to be a constant and satisfies $0<\zeta\leq 1$. However, the deception attack handled in this article is a time-varying signal that makes control design complicated. On the other hand, from the mathematical expression form, faults and attacks are essentially the same, but the malicious attackers will choose the optimal attack strategies, resulting in undetectable attack signals in practice. As a direct result, only the compromised system output $\breve{x}_1$ can be used by the designer. Thus, the proposed control strategies in [21] and [22] are no longer applicable, and a new controller design strategy using the compromised error surfaces needs to be considered.

In addition, the hysteretic quantizer considered in this article is given by [23]

$$q(u)=\begin{cases}u_i\mathrm{sgn}(u), & \frac{u_i}{1+\delta}<|u|\leq u_i,\dot{u}<0,\text{ or}\\ & u_i<|u|\leq\frac{u_i}{1-\delta},\dot{u}>0\\ u_i(1+\delta)\mathrm{sgn}(u), & u_i<|u|\leq\frac{u_i}{1-\delta},\dot{u}<0,\text{ or}\\ & \frac{u_i}{1-\delta}<|u|\leq\frac{u_i(1+\delta)}{1-\delta},\dot{u}>0\\ 0, & 0\leq|u|<\frac{u_{\min}}{1+\delta},\dot{u}<0,\text{ or}\\ & \frac{u_{\min}}{1+\delta}\leq|u|<u_{\min},\dot{u}>0\\ q(u(t^-)) & \dot{u}=0\end{cases}\tag{2}$$

where $u_i=\mu^{1-i}u_{\min}$, $i=1,2,\ldots$, and $u_{\min}>0$ denotes the size of the dead zone for $q(u)$ and $0<\mu<1$, $\delta=(1-\mu)/(1+\mu)$, and μ is a measure of quantization density. The map of the hysteretic quantizer $q(u)$ for $u>0$ is shown in Fig. 3. It is noted that if the rate of change of $\dot{u}$ is positive (i.e., u is increasing) and its absolute value $|u|$ is less than the upper limit of the dead band $\frac{u_{\min}}{1+\delta}$, i.e., $|u|<\frac{u_{\min}}{1+\delta}$, the output of quantizer is 0. The purpose of this rule is to quantify the input signal to 0 when the direction of change is positive and the amplitude is small, to maintain the stability of the dead zone and minimize quantization errors near zero.

In order to propose a suitable control scheme, the hysteretic quantizer $q(u)$ can be decomposed into a linear part and a nonlinear part as follows:

$$q(u)=u+d.\tag{3}$$

Lemma 1: The nonlinearity d satisfies the following inequalities:

$$d^2\leq\delta^2u^2\quad\forall|u|\geq u_{\min}\tag{4}$$

$$d^2\leq u_{\min}^2\quad\forall|u|\leq u_{\min}.\tag{5}$$

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Proof: The proof is given in [23]. ■

Lemma 2 (see [24]): On $[0,\infty)$, define two smooth functions $V(\cdot)$ and $\kappa(\cdot)$ with $V(t)\geq 0$ and a smooth Nussbaum-type function $N(\cdot)$. If the following inequality holds:

$$\dot{V}(t)\leq D+\int_0^t\left(\beta(t)N(\kappa)+1\right)\dot{\kappa}d\tau \tag{6}$$

where D represents a suitable positive constant, then $V(t)$, $\kappa(t)$, and $\int_0^t(\beta(t)N(\kappa)+1)\dot{\kappa}d\tau$ are bounded over $[0,\infty)$.

Lemma 3 (see [10]): For any real-valued continuous function $s(x,y)$, there exist smooth functions $s_1(x)\geq 1$ and $s_2(y)\geq 1$ so that the inequality $|s(x,y)|\leq s_1(x)s_2(y)$ holds.

Lemma 4: For any given $\varpi\in R^m$, there exists $\sigma>0$, such that

$$0\leq\|\varpi\|-\frac{\|\varpi\|^2}{\sqrt{\|\varpi\|^2+\sigma^2}}\leq\sigma. \tag{7}$$

III. Design of the Resilient Adaptive Controller Based on K-Filters

A. K-Filter Design

In this article, only the output x_1 can be measured in system (1); therefore, a state observer needs to be developed to realize the estimation of unknown state x_i, $i=2,\ldots,n$. However, unlike the general output feedback control method [25], the output of the system is corrupted due to the existence of deception attacks. Thus, only the false state $\check{x}_1$ can be used to design the observer, making the observer design challenging.

In order to avoid the effect of unknown attack signal $\omega_s(t,x_1)$ under deception attack, a set of new K-filters for estimating $n-1$ unmeasurable states $x_2,\ldots,x_n$ using only false output $\check{x}_1$ is constructed as follows:

$$\dot{\xi}_0=A_0\xi_0+H_0\check{x}_1,\quad \xi_0\in R^n \tag{8}$$

$$\dot{\xi}_1=A_0\xi_1+\phi(\check{x}_1),\quad \xi_1\in R^{n\times p} \tag{9}$$

$$\dot{\xi}_2=A_0\xi_2+\psi(\check{x}_1),\quad \xi_2\in R^n \tag{10}$$

$$\dot{\chi}_a=A_0\chi_a+e_{n,n}\Delta(\check{x}_1),\quad \chi_a\in R^n \tag{11}$$

$$\dot{\chi}=A_0\chi+e_{n,n}q(u),\quad \chi\in R^n \tag{12}$$

where $A_0=\begin{bmatrix}-k_1 & 1 & & \\ -k_2 & 0 & 1 & \\ \vdots & \vdots & & \ddots \\ -k_{n-1} & 0 & \ldots & 0 & 1 \\ -k_n & 0 & \cdots & 0 & 0\end{bmatrix}\in R^{n\times n}$ with $k=[k_1,\ldots,k_n]^T\in R^n$ is chosen such that the matrix A_0 is a strict Hurwitz matrix. $H_0=[1,1,\ldots,1]^T\in R^n$, $\phi(\cdot)=[\phi_1(\cdot),\phi_2(\cdot),\ldots,\phi_n(\cdot)]^T\in R^{n\times p}$, $\psi(\cdot)=[\psi_1(\cdot),\psi_2(\cdot),\ldots,\psi_n(\cdot)]^T\in R^n$, and $e_{n,n}=[0,\ldots,1]^T\in R^n$.

Using the mean value theory [26], it naturally holds that $\phi(\check{x}_1)=\phi(x_1)+o_1(1-\lambda)H_0\check{x}_1$, $\psi(\check{x}_1)=\psi(x_1)+o_2(1-\lambda)H_0\check{x}_1$, and $\Delta(\check{x}_1)=\Delta(x_1)+o_3(1-\lambda)H_0\check{x}_1$, where $o_1=\text{diag}\{o_{1,j}\}$ with $o_{1,j}=[o_{1,j1},\ldots,o_{1,jp}]\in R^p$, $o_2=\text{diag}\{o_{2,j}\}$, and $o_3=[o_{3,1},0,\ldots,0]\in R^n$ are all the unknown derivatives, $j=1,\ldots,n$.

Define $h_j=A_0^j\chi_a$ and $\eta_j=A_0^j\chi$, $j=0,1,\ldots,m$, by using [27, eqs. (8.22) and (8.23)]; then, we have $\dot{h}_j=A_0^j\dot{\chi}_a=A_0^j(A_0\chi_a+e_{n,n}\Delta(\check{x}_1))=A_0h_j+e_{n,n-j}\Delta(\check{x}_1)$ and $\dot{\eta}_j=A_0^j\dot{\chi}=A_0^j(A_0\chi+e_{n,n}q(u))=A_0\eta_j+e_{n,n-j}q(u)$, where $A_0^je_{n,n}=e_{n,n-j}$, $j=0,1,\ldots,n$.

Based on these designed filters, we introduce

$$\hat{x}=\Xi\xi_0+\xi_1\theta+\xi_2+\sum_{j=0}^m b_j\eta_j+w_a(t)\sum_{j=0}^m b_jh_j \tag{13}$$

where $\hat{x}=[\hat{x}_1,\hat{x}_2,\ldots,\hat{x}_n]^T$ is the state estimation of x, $\Xi=k\lambda-o_1\theta(1-\lambda)-o_2(1-\lambda)-\sum_{j=0}^m e_{n,n-j}b_jw_a(t)o_3(1-\lambda)$, and $k=\text{diag}\{k_1,\ldots,k_n\}$.

From (8)–(13), we have

$$\begin{aligned}\dot{\hat{x}}=&\,\Xi(A_0\xi_0+H_0\check{x}_1)+\dot{\Xi}\xi_0+A_0\xi_1\theta+\phi(\check{x}_1)\theta+A_0\xi_2\\&+\psi(\check{x}_1)+\sum_{j=0}^m b_jA_0\eta_j+b_je_{n,n-j}q(u)+\dot{w}_a\sum_{j=0}^m b_jh_j\\&+w_a\sum_{j=0}^m b_jA_0h_j+b_je_{n,n-j}w_a\Delta(\check{x}_1)\\=&\,A_0\hat{x}+\begin{bmatrix}0_{(r-1)\times 1}\\ b\end{bmatrix}[q(u)+w_a(t)\Delta(x_1)]+k\lambda H_0\check{x}_1\\&+\dot{\Xi}\xi_0+\dot{w}_a(t)\sum_{j=0}^m b_jh_j+\phi(y)\theta+\psi(y).\end{aligned} \tag{14}$$

Defining the estimation error as $\varepsilon=x-\hat{x}$, then by (14), after some mathematical derivations, one obtains

$$\dot{\varepsilon}=A_0\varepsilon-\dot{\Xi}\xi_0-\dot{w}_a(t)\sum_{j=0}^m b_jh_j. \tag{15}$$

By (1) and (13), it follows that:

$$\begin{aligned}\dot{y}=&\,\varepsilon_2+\Xi_2\xi_0+\xi_{1,2}\theta+\xi_{2,2}+b_m\eta_{m,2}+w^T\Theta\\&+\sum_{j=0}^m w_ab_jh_{j,2}+\phi_1(y)\theta+\psi_1(y)\end{aligned} \tag{16}$$

where $w=[0,\eta_{m-1,2},\ldots,\eta_{0,2}]^T$, $\Theta=[b_m,b_{m-1},\ldots,b_0]^T$, and $\eta_{j,2}$, ε_2, Ξ_2, $\xi_{1,2}$, $\xi_{2,2}$, and $h_{j,2}$ are all the second row of η_j, ε, Ξ, ξ_1, ξ_2, and h_j, $j=0,1,\ldots,m$.

Remark 3: When the attack occurs, the expected control performance cannot be achieved. Under the circumstances, how to design a suitable state estimator is still a challenging work. Here, a new form of K-filters in (8)–(12) with the false output and a virtual estimate term in (13) is proposed such that the attack signal can be effectively compensated.

B. Design of the Resilient Adaptive Controller

Now, we replace (1) with a new system, whose states depend on filters (8)–(12) and thus are available for control design under deception attacks

$$\dot{y}=\varepsilon_2+\Xi_2\xi_0+\xi_{1,2}\theta+\xi_{2,2}+b_m\eta_{m,2}+w^T\Theta$$

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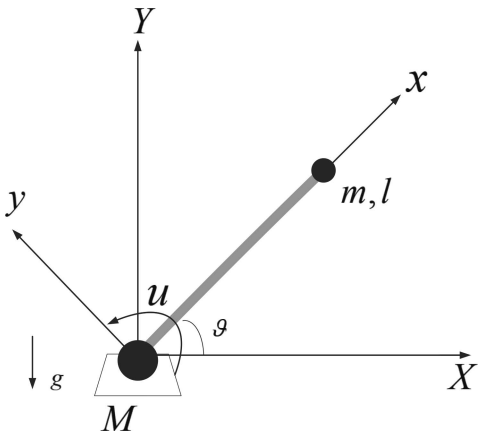


Fig. 4. Framework of a single-link robot.

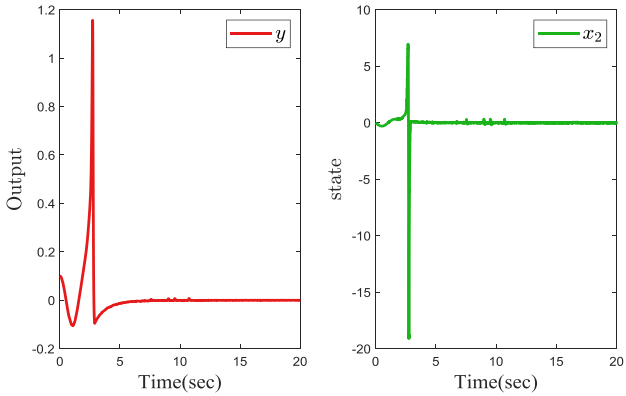


Fig. 5. Trajectories of y and x_2.

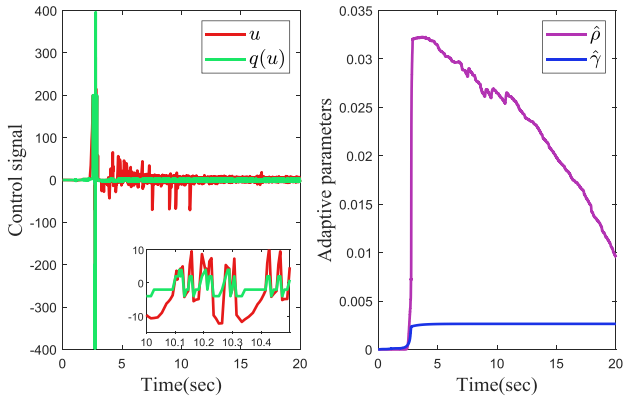


Fig. 6. Response curves of u, $q(u)$, $\hat{\rho}$, and $\hat{\gamma}$.

$$+\sum_{j=0}^{m} w_a b_j h_{j,2} + \phi_1(y)\theta + \psi_1(y)$$

$$\dot{\eta}_{m,q} = \eta_{m,q+1} - k_q \eta_{m,1}, \quad q = 1, \ldots, r-1$$
$$\dot{\eta}_{m,r} = \eta_{m,r+1} - k_r \eta_{m,1} + q(u) \tag{17}$$

where $\eta_{m,q},\ q = 1, \ldots, r$, is the qth element of η_m. Before proceeding to the recursive design, a new coordinate transformation of the error is introduced

$$z_1 = x_1, \quad \check{z}_1 = \check{x}_1$$
$$z_q = \eta_{m,q} - \alpha_{q-1}, \quad q = 2, \ldots, r \tag{18}$$

where α_{q-1} is the virtual control signals to be specified later.

Remark 4: It should be pointed out that the error surface $z_1 = x_1$ in (18) cannot be applied to the process of controller design, which is only appeared in the process of achieving the stabilization of the closed-loop system. This is a sharp contrast between the general recursive design and the presented recursive design strategy for nonlinear systems subject to deception attacks.

Let us choose the control law as

$$u = -\frac{z_r \alpha_r^2}{(1-\delta)\sqrt{z_r^2 \alpha_r^2 + \sigma^2}} - \frac{u_{\min} z_r}{(1-\delta)\sqrt{z_r^2 + \sigma^2}} \tag{19}$$

with

$$\alpha_1 = N(\kappa)\Phi, \dot{\kappa} = \check{z}_1 \Phi, \Phi = \bar{s}_1 \check{z}_1 + \hat{\gamma} v_1 \tag{20}$$

$$\alpha_2 = -\left(s_2 + \frac{1}{4}\right) z_2 - \hat{\rho} v_2 + \Lambda_2 + k_2 \eta_{m,1} \tag{21}$$

$$\alpha_3 = -s_3 z_3 - z_2 + k_3 \eta_{m,1} + \Lambda_3 + \frac{\partial \alpha_2}{\partial \hat{\rho}} \delta_2 \varsigma_3 \tag{22}$$

$$\alpha_q = -s_q z_q - z_{q-1} + k_q \eta_{m,1} + \Lambda_q + \frac{\partial \alpha_{q-1}}{\partial \hat{\rho}} \varsigma_q + \delta_2 \left(\sum_{k=3}^{q-1} z_k \frac{\partial \alpha_{k-1}}{\partial \hat{\rho}} z_q v_q \right) \tag{23}$$

$$\alpha_r = s_r z_r + z_{r-1} + \eta_{m,r+1} - k_r \eta_{m,1} - \Lambda_r - \sum_{l=3}^{r-1} z_l \frac{\partial \alpha_{l-1}}{\partial \hat{\rho}} \delta_2 z_r^2 v_r - \frac{\partial \alpha_{r-1}}{\partial \hat{\rho}} \varsigma_r \tag{24}$$

where $\varsigma_2 = \delta_2 z_2 v_2$, $\varsigma_3 = \varsigma_2 + \delta_2 z_3 v_3$, and $\varsigma_q = \varsigma_{q-1} + \delta_2 z_q^2 v_q$ are the tuning functions, α_q, $q = 4, \ldots, r-1$, ς_q, $q = 4, \ldots, r$, and $\bar{s}_1$, s_2, …, s_r are the positive design parameters, and

$$\begin{aligned}\Lambda_q = {} & \frac{\partial \alpha_{q-1}}{\partial \hat{\gamma}} \dot{\hat{\gamma}} + \frac{\partial \alpha_{q-1}}{\partial \sigma} \dot{\sigma} + \frac{\partial \alpha_{q-1}}{\partial \xi_1} (A_0 \xi_1 + \phi(\check{z}_1)) \\ & + \frac{\partial \alpha_{q-1}}{\partial \xi_2} (A_0 \xi_2 + \psi(\check{z}_1)) + \frac{\partial \alpha_{q-1}}{\partial \xi_0} (A_0 \xi_0 + H_0 \check{z}_1) \\ & + \sum_{j=1}^{m+q-2} \frac{\partial \alpha_{q-1}}{\partial \chi_j} (-k_j \chi_1 + \chi_{j+1}) + \frac{\partial \alpha_{q-1}}{\partial \varrho} \dot{\varrho} \\ & + \frac{\partial \alpha_{q-1}}{\partial \kappa} \dot{\kappa} + \sum_{j=0}^{m} \frac{\partial \alpha_{q-1}}{\partial h_{j,2}} \dot{h}_{j,2}, \quad q = 3, \ldots, r. \end{aligned} \tag{25}$$

Choose the following adaptive laws:

$$\dot{\hat{\gamma}} = \delta_1 \check{z}_1 v_1, \dot{\hat{\rho}} = \varsigma_r, \varsigma_r = \varsigma_{r-1} + \delta_2 z_r^2 v_r \tag{26}$$

where δ_1 and δ_2 are the positive design parameters, and $\hat{\gamma}$ and $\hat{\rho}$ are the estimate of γ and ρ, respectively, which will be defined later.

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Now, the main results in this article can be concluded as follows.

Theorem 1: Consider a class of nonlinear systems (1) suffering from unknown sensor and actuator attacks. Suppose that Assumptions 1 and 2 hold; if the filters (8)–(12), the resilient adaptive controller (19), and the adaptive update laws (26) are applied, then all the signals of the closed-loop systems are globally uniformly bounded, and the output y of the system converges to zero, i.e., $\lim_{t\to\infty} y(t) = 0$.

Proof: See the Appendix. ■

Remark 5: Inspired by Yu and Lin [20], by proposing a novel transformation of the final controller in (19) rather than $-\alpha_r$ in (24), where α_r is obtained via the backstepping design approach, the effect of the nonlinearity that exists in the quantizer on the system stability can be effectively eliminated, and the asymptotic stability under deception attacks is achieved in this article.

IV. Simulation Studies

The stabilization problem of a single-link robot (see Fig. 4) under sensor and actuator attacks is simulated in this section, which can be described by [28]

$$M\ddot{\vartheta} + \frac{1}{2}mgl\sin(\vartheta) = u, y = \vartheta. \tag{27}$$

In the simulation, we consider the hysteretic quantizer in (2). Let $x_1 = \vartheta$ and $x_2 = \dot{\vartheta}$; then, the system model can be rewritten as the output-feedback form

$$\begin{aligned} \dot{x}_1 &= x_2 \\ \dot{x}_2 &= \theta\phi_2(y) + b_0[q(u) + \rho_a(t, x_1)] \\ y &= x_1 \end{aligned} \tag{28}$$

where the unknown parameters are defined as $\theta = -\frac{1}{2M}mgl$, $b_0 = \frac{1}{M} > 0$. $M = 0.5$, $g = 9.8$, $m = 1$, $l = 1$, and $\phi_2(y) = \sin(y)$. The physical significance for related parameters can refer to [28]. In simulation, the initial states are selected as $x_1(0) = 0.1$ and $x_2(0) = 0$. The attack are chosen as $\zeta(t) = 1 + 0.25\cos(t)$ and $\rho_a(t, x_1) = x_1\sin(1.5t)$. Quantizer parameters are chosen as $u_{\min} = 0.02$ and $\mu = 0.01$. The design parameters are chosen as $\bar{s}_1 = 2$, $s_2 = 5$, $k_1 = 0.5$, $r_1 = 0.5$, $r_2 = 2$, $\delta_1 = 0.001$, and $\delta_2 = 0.005$.

Simulation results are displayed in Figs. 5 and 6, respectively. Fig. 5 shows output y and state x_2. Fig. 6 gives the control input signal u and the estimation parameters $\hat{\rho}$ and $\hat{\gamma}$. From these figures, we can clearly obtain that the proposed resilient adaptive control algorithm ensures that the output of the system converges to zero regardless of sensor and actuator attacks, and all the signals are bounded.

V. Conclusion

This study has investigated an adaptive resilient quantized control problem for nonlinear CPSs against sensor and actuator deception attacks. The innovation of this article is that only the compromised output is transmitted for the feedback, and a set of new K-filters is proposed to reconstruct parts of unmeasured states. A new coordinate transformation involving the compromised output and estimated state variables is constructed, with which a backstepping control design strategy capable of achieving steady-state performance for the nonlinear CPSs under deception attacks is developed. It is shown that the output of the system converges to zero, and all the signals in the closed-loop system are globally uniformly bounded. Nevertheless, as a common disadvantage of the Nussbaum function approach, our scheme may suffer from system oscillation. In future work, we will focus on the improvement of the design to avoid system oscillation.

Appendix

This appendix provides the proof of state boundedness of the system. Now, the backstepping design procedure is presented step by step.

Step 1: The Lyapunov function V_1 is chosen as

$$V_1 = \frac{\lambda^2}{2\lambda_M}z_1^2 + \frac{1}{2\delta_1}\tilde{\gamma}^2 \tag{29}$$

where $\lambda = (1+\zeta)^{-1}$; by recalling Assumption 2, we have $\lambda_m \le |\lambda| \le \lambda_M$ and $|\dot{\lambda}| \le \bar{\dot{\lambda}}$, where λ_m, λ_M, and $\bar{\dot{\lambda}}$ are unknown positive constants. According to the definition of λ, the formula $x_1 = \lambda\check{x}_1$ holds. In addition, $\delta_1 > 0$ is a design parameter and $\tilde{\gamma} = \gamma - \hat{\gamma}$, where $\hat{\gamma}$ is the estimate of γ, which will be defined later.

From (13)–(18), the derivative of V_1 is given as

$$\begin{aligned} \dot{V}_1 = {} & \frac{\lambda\dot{\lambda}}{\lambda_M}z_1^2 + \frac{\lambda^2 z_1}{\lambda_M}(\varepsilon_{a,2} - \varepsilon_{b,2} - \varepsilon_{c,2} + \Xi_2\xi_0 + \xi_{1,2}\theta + \xi_{2,2} \\ & + b_m z_2 + b_m\alpha_1 + w^T\Theta + \sum_{j=0}^{m} w_a b_j h_{j,2} + \phi_1(y)\theta \\ & + \psi_1(y)) - \frac{1}{\delta_1}\tilde{\gamma}\dot{\hat{\gamma}} \end{aligned} \tag{30}$$

where $\varepsilon_2 = \varepsilon_{a,2} - \varepsilon_{b,2} - \varepsilon_{c,2}$, and $\varepsilon_{a,2}$, $\varepsilon_{b,2}$, and $\varepsilon_{c,2}$ are the second elements of ε_a, ε_b, and ε_c, respectively. $\varepsilon_a = e^{A_0 t}\varepsilon_0$, $\varepsilon_b = \int_0^t e^{A_0(t-\tau)}\dot{\Xi}(\tau)\xi_0(\tau)d\tau$, and $\varepsilon_c = \sum_{j=0}^{m}\int_0^t e^{A_0(t-\tau)}b_j\dot{w}_a(\tau)h_j(\tau)d\tau$. Then, from (15), it is easy to obtain that $\varepsilon = \varepsilon_a - \varepsilon_b - \varepsilon_c$. Furthermore, we have

$$\begin{aligned} |\varepsilon_{b,2}| + |\varepsilon_{c,2}| \le {} & \int_0^t \|e_{n,2}^T e^{A_0(t-\tau)}\|\|\dot{\Xi}(\tau)\|\|\xi_0(\tau)\|d\tau \\ & + \sum_{j=0}^{m}\int_0^t \|e_{n,2}^T e^{A_0(t-\tau)}\||b_j\dot{w}_a(\tau)|\|h_j(\tau)\|d\tau \\ \le {} & \gamma_s\varrho \end{aligned} \tag{31}$$

where $\gamma_b = \sup\|\dot{\Xi}(\tau)\|$, $\gamma_c = \max\{|b_j|\bar{w}_a, j = 0, \ldots, m\}$, and $\gamma_s = \max\{\gamma_b, \gamma_c\}$. ϱ is the auxiliary variable, which is designed as $\dot{\varrho} = -r_1\varrho + r_2(\|\xi_0(\tau)\| + \sum_{j=0}^{m}\|h_j(\tau)\|)$ with $\varrho(0) \ge 0$, and r_1 and r_2 are the positive constants.

According to Lemma 3, as for $\phi_1(y)$, there exist $h_\phi(\cdot)$ and $d_1(\cdot)$ such that $\|\phi_1(y)\| = \|\phi_1(\lambda\check{x}_1)\| \le h_\phi(\lambda)d_1(\check{x}_1) \le \lambda_\phi d_1(\check{x}_1)$, where λ_ϕ is the unknown constant bound of $h_\phi(\lambda)$ and $d_1(\check{x}_1)$ is a known smooth function. Therefore, by using

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Lemma 4, we can get

$$\frac{\lambda^2 z_1}{\lambda_M}\phi_1(y)\theta \le \gamma_e(\sigma + v_{e,1}\check{z}_1) \tag{32}$$

where $\gamma_e = \lambda_M^2\lambda_\phi\|\theta\|$ and $v_{e,1} = \frac{\check{z}_1 d_1^2(\check{z}_1)}{\sqrt{[\check{z}_1 d_1(\check{z}_1)]^2+\sigma^2}}$.

Similarly, using Young's inequality and Lemma 4, we have

$$\dot{V}_1 \le \check{z}_1\gamma v_1 + \frac{1}{4}z_2^2 + \frac{\lambda^2 b_m}{\lambda_M}z_1\alpha_1 + \gamma\sigma_s + \frac{1}{4}\varepsilon_a^T\varepsilon_a - \frac{1}{\delta_1}\tilde{\gamma}\dot{\hat{\gamma}} \tag{33}$$

where $\sigma_s = (7+m)\sigma$, $\sigma(t)$ is a positive uniformly continuous and bounded function for $t \ge 0$, and the inequality $\lim_{t\to\infty}\int_0^t \sigma(\tau)d\tau \le \bar{\sigma} < +\infty$ holds, where $\bar{\sigma} > 0$ is a constant. $\gamma = \max\{\gamma_a, \gamma_b, \gamma_c, \gamma_d, \gamma_e, \gamma_f, \gamma_g, \gamma_h, \gamma_p, \gamma_w, \gamma_l\}$, $v_1 = v_{a,1} + v_{b,1} + v_{c,1} + v_{d,1} + v_{e,1} + v_{f,1} + v_{g,1} + v_{h,1} + v_{p,1} + v_{w,1} + v_{l,1}$, $\gamma_a = \lambda_M^4$, $\gamma_b = \lambda_M^2\|\theta\|$, γ_c is the upper bound of $\|\lambda_M^2\Xi_1\|$, $\gamma_d = \lambda_M^2$, $\gamma_f = \lambda_M^2\lambda_\psi$, $\gamma_g = \lambda_M^2\gamma_s$, $\gamma_h = \lambda_M^2\bar{\dot{\lambda}}$, $\gamma_p = b_m^2\lambda_M^4$, $\gamma_w = \|\lambda_M^2\Theta\|$, and $\gamma_l = [\gamma_{l0}, \ldots, \gamma_{lm}]^T$ with $\gamma_{lj} = \lambda_M^2\bar{w}_a|b_j|$, $j = 0, \ldots, m$. $v_{a,1} = \check{x}_1$, $v_{b,1} = \frac{\check{z}_1\xi_{1,2}^T\xi_{1,2}}{\sqrt{\check{z}_1^2\xi_{1,2}^T\xi_{1,2}+\sigma^2}}$, $v_{c,1} = \frac{\check{z}_1\xi_0^T\xi_0}{\sqrt{\check{z}_1^2\xi_0^T\xi_0+\sigma^2}}$, $v_{d,1} = \frac{\check{z}_1\xi_{2,2}^2}{\sqrt{(\check{z}_1\xi_{2,2})^2+\sigma^2}}$, $v_{f,1} = \frac{\check{z}_1 q_1^2(\check{z}_1)}{\sqrt{[\check{z}_1 q_1(\check{z}_1)]^2+\sigma^2}}$, $v_{g,1} = \frac{\varrho^2\check{z}_1}{\sqrt{(\varrho\check{z}_1)^2+\sigma^2}}$, $v_{h,1} = \check{z}_1$, $v_{p,1} = \check{z}_1$, $v_{w,1} = \frac{\check{z}_1 w^T w}{\sqrt{\check{z}_1^2 w^T w+\sigma^2}}$, and $v_{l,1} = \left[\frac{\check{z}_1 h_{0,2}^2}{\sqrt{(\check{z}_1 h_{0,2})^2+\sigma^2}}, \ldots, \frac{\check{z}_1 h_{m,2}^2}{\sqrt{(\check{z}_1 h_{m,2})^2+\sigma^2}}\right]^T$.

By substituting (20) into (33), one can obtain

$$\dot{V}_1 \le -s_1 z_1^2 + \tilde{\gamma}\check{z}_1 v_1 + (\beta(t)N+1)\dot{\kappa} - \frac{1}{\delta_1}\tilde{\gamma}\dot{\hat{\gamma}} + \frac{1}{4}z_2^2 + \gamma\sigma_s + \frac{1}{4}\varepsilon_a^T\varepsilon_a \tag{34}$$

where $s_1 = \bar{s}_1\lambda_M^{-2}$ and $\beta(t) = \lambda_M^{-1}\lambda^3 b_m$.

Step 2: Differentiating z_2, we obtain

$$\dot{z}_2 = z_3 + \alpha_2 - k_2\eta_{m,1} - \frac{1}{\lambda}\frac{\partial\alpha_1}{\partial\check{z}_1}\Bigg(\varepsilon_{a,2} - \varepsilon_{b,2} - \varepsilon_{c,2} + \Xi_2\xi_0 + \xi_{1,2}\theta + \xi_{2,2} + b_m\eta_{m,2} + w^T\Theta + \phi_1(y)\theta + \psi_1(y) + \sum_{j=0}^{m} w_a b_j h_{j,2}\Bigg) + \frac{\dot{\lambda}}{\lambda}\frac{\partial\alpha_1}{\partial\check{z}_1}\check{z}_1 - \Lambda_2 \tag{35}$$

where $\Lambda_2 = \frac{\partial\alpha_1}{\partial\hat{\gamma}}\dot{\hat{\gamma}} + \frac{\partial\alpha_1}{\partial\sigma}\dot{\sigma} + \frac{\partial\alpha_1}{\partial\xi_1}(A_0\xi_1 + \phi(\check{z}_1)) + \frac{\partial\alpha_1}{\partial\xi_2}(A_0\xi_2 + \psi(\check{z}_1)) + \frac{\partial\alpha_1}{\partial\xi_0}(A_0\xi_0 + H_0\check{z}_1) + \sum_{j=1}^{m}\frac{\partial\alpha_1}{\partial\chi_j}(-k_j\chi_1 + \chi_{j+1}) + \frac{\partial\alpha_1}{\partial\varrho}\dot{\varrho} + \frac{\partial\alpha_1}{\partial\kappa}\dot{\kappa} + \sum_{j=0}^{m}\frac{\partial\alpha_1}{\partial h_{j,2}}\dot{h}_{j,2}$.

Choose the second Lyapunov function candidate as $V_2 = V_1 + \frac{1}{2}z_2^2 + \frac{1}{2\delta_2}\tilde{\rho}^2$, where $\delta_2 > 0$ is a design parameter and $\tilde{\rho} = \rho - \hat{\rho}$; $\hat{\rho}$ is the estimate of ρ.

In light of (35), the derivative of V_2 is computed as

$$\dot{V}_2 \le \dot{V}_1 + z_2\rho v_2 + z_2(z_3 + \alpha_2) + \rho\sigma_p + \frac{1}{4}\varepsilon_a^T\varepsilon_a - \frac{1}{\delta_2}\tilde{\rho}\dot{\hat{\rho}} \tag{36}$$

where $\rho = \max\{\rho_a, \rho_b, \rho_c, \rho_d, \rho_e, \rho_f, \rho_g, \rho_h, \rho_p, \rho_w, \rho_l\}$, $\rho_a = \lambda_m^{-2}$, $\rho_b = \lambda_m^{-1}|\theta|$, ρ_c is the upper bound of $\|\lambda_m^{-1}\Xi_1\|$, $\rho_d = \lambda_m^{-1}$, $\rho_e = \lambda_m^{-1}\lambda_\phi\|\theta\|$, $\rho_f = \lambda_m^{-1}\lambda_\psi$, $\rho_g = \lambda_m^{-1}\gamma_s$, $\rho_h = \lambda_m^{-1}\bar{\dot{\lambda}}$, $\rho_p = \lambda_m^{-1}b_m$, $\rho_w = \|\lambda_m^{-1}\Theta\|$, and $\rho_l = [\rho_{l0}, \ldots, \rho_{lm}]^T$ with $\rho_{lj} = \lambda_m^{-1}\bar{w}_a|b_j|$, $j = 0, \ldots, m$. For $q = 2, \ldots, r$, $v_q = v_{a,q} + v_{b,q} + v_{c,q} + v_{d,q} + v_{e,q} + v_{f,q} + v_{g,q} + v_{h,q} + v_{p,q} + v_{w,q} + v_{l,q}$, $v_{a,q} = z_q\left(\frac{\partial\alpha_1}{\partial\check{z}_1}\right)^2$, $v_{b,q} = \frac{z_q(\frac{\partial\alpha_1}{\partial\check{z}_1})^2\xi_{1,2}^T\xi_{1,2}}{\sqrt{(z_q\frac{\partial\alpha_1}{\partial\check{z}_1}\xi_{1,2})^2+\sigma^2}}$, $v_{c,q} = \frac{z_q\left(\frac{\partial\alpha_1}{\partial\check{z}_1}\right)^2\xi_0^T\xi_0}{\sqrt{z_q^2\left(\frac{\partial\alpha_1}{\partial\check{z}_1}\right)^2\xi_0^T\xi_0+\sigma^2}}$, $v_{d,q} = \frac{z_q\left(\frac{\partial\alpha_1}{\partial\check{z}_1}\right)^2\xi_{2,2}^2}{\sqrt{\left(z_q\frac{\partial\alpha_1}{\partial\check{z}_1}\xi_{2,2}\right)^2+\sigma^2}}$, $v_{e,q} = \frac{z_q\left(\frac{\partial\alpha_1}{\partial\check{z}_1}\right)^2 d_1^2(\check{z}_1)}{\sqrt{\left[z_q\frac{\partial\alpha_1}{\partial\check{z}_1}d_1(\check{z}_1)\right]^2+\sigma^2}}$, $v_{f,q} = \frac{z_q\left(\frac{\partial\alpha_1}{\partial\check{z}_1}\right)^2 q_1^2(\check{z}_1)}{\sqrt{\left[z_q\frac{\partial\alpha_1}{\partial\check{z}_1}q_1(\check{z}_1)\right]^2+\sigma^2}}$, $v_{g,q} = \frac{z_q\left(\frac{\partial\alpha_1}{\partial\check{z}_1}\right)^2\varrho^2}{\sqrt{\left(z_q\frac{\partial\alpha_1}{\partial\check{z}_1}\varrho\right)^2+\sigma^2}}$, $v_{h,q} = \frac{z_q\left(\frac{\partial\alpha_1}{\partial\check{z}_1}\right)^2\check{z}_1^2}{\sqrt{\left(z_q\frac{\partial\alpha_1}{\partial\check{z}_1}\check{z}_1\right)^2+\sigma^2}}$, $v_{p,q} = \frac{z_q\left(\frac{\partial\alpha_1}{\partial\check{z}_1}\right)^2\eta_{m,2}^2}{\sqrt{(z_q\frac{\partial\alpha_1}{\partial\check{z}_1}\eta_{m,2})^2+\sigma^2}}$, $v_{w,q} = \frac{z_q\left(\frac{\partial\alpha_1}{\partial\check{z}_1}\right)^2 w^T w}{\sqrt{z_q^2\left(\frac{\partial\alpha_1}{\partial\check{z}_1}\right)^2 w^T w+\sigma^2}}$, and $v_{l,q} = \left[\frac{z_q(\frac{\partial\alpha_1}{\partial\check{z}_1})^2 h_{0,2}^2}{\sqrt{(z_q(\frac{\partial\alpha_1}{\partial\check{z}_1})h_{0,2})^2+\sigma^2}}, \ldots, \frac{z_q\left(\frac{\partial\alpha_1}{\partial\check{z}_1}\right)^2 h_{m,2}^2}{\sqrt{(z_q\left(\frac{\partial\alpha_1}{\partial\check{z}_1}\right)h_{m,2})^2+\sigma^2}}\right]^T$.

Substituting (21) into (36) yields

$$\dot{V}_2 \le \sum_{l=1}^{2} -s_l z_l^2 + \tilde{\gamma}\check{z}_1 v_1 + (\beta(t)N+1)\dot{\kappa} - \frac{1}{\delta_1}\tilde{\gamma}\dot{\hat{\gamma}} + \gamma\sigma_s + \frac{1}{2}\varepsilon_a^T\varepsilon_a + \frac{1}{\delta_2}\tilde{\rho}(\varsigma_2 - \dot{\hat{\rho}}) + z_2 z_3 + \rho\sigma_p \tag{37}$$

where $\sigma_p = (9+m)\sigma$.

Step q ($3 \le q \le r-1$): By following the ideas in the aforementioned two steps, the virtual control laws are chosen as (22) and (23), and with the help of the Lyapunov function $V_q = V_{q-1} + \frac{1}{2}z_q^2$, it follows that:

$$\dot{V}_q \le \sum_{l=1}^{q} -s_l z_l^2 + z_q z_{q+1} + \tilde{\gamma}\check{z}_1 v_1 + (\beta(t)N+1)\dot{\kappa} - \frac{1}{\delta_1}\tilde{\gamma}\dot{\hat{\gamma}} + \gamma\sigma_s + \frac{q}{4}\varepsilon_a^T\varepsilon_a - \sum_{l=3}^{q} z_l\frac{\partial\alpha_{l-1}}{\partial\hat{\rho}}(\dot{\hat{\rho}} - \varsigma_q) + \frac{1}{\delta_2}\tilde{\rho}(\varsigma_q - \dot{\hat{\rho}}) - \frac{q}{4}\varepsilon_a^T\varepsilon_a + (q-1)\rho\sigma_p. \tag{38}$$

Step r: The final controller and adaptive laws are recursively derived at the final step r, as summarized in (19) and (26). From (18) and using (24), the derivative of z_r is described as $\dot{z}_r = -s_r z_r - z_{r-1} + q(u) + \alpha_r$. Considering the Lyapunov function $V_r = V_{r-1} + \frac{1}{2}z_r^2$, from (38) and (26), we have

$$\dot{V}_r \le \sum_{l=1}^{r} -s_l z_l^2 + (\beta(t)N+1)\dot{\kappa} - +\gamma\sigma_s + (r-1)\rho\sigma_p + z_r\alpha_r + z_r(u+d) - \frac{r}{4}\varepsilon_a^T\varepsilon_a. \tag{39}$$

Noting from (19) that $z_r u < 0$ and considering Lemma 4, we have

$$z_r(u+d) \le z_r u + \delta|z_r u| + u_{\min}|z_r|$$

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$$\leq (1-\delta)z_r u + u_{\min}\left(\frac{z_r^2}{\sqrt{z_r^2+\sigma^2}}+\sigma\right)$$
$$\leq -\frac{z_r^2\alpha_r^2}{\sqrt{z_r^2\alpha_r^2+\sigma^2}} + u_{\min}\sigma$$
$$\leq \sigma - z_r\alpha_r + u_{\min}\sigma. \tag{40}$$

Then, we get

$$\dot{V}_r \leq \sum_{l=1}^{r} -s_l z_l^2 + (\beta(t)N+1)\dot{\kappa} + \gamma\sigma_s + (r-1)\rho\sigma_p + (u_{\min}+1)\sigma - \frac{r}{4}\varepsilon_a^T\varepsilon_a. \tag{41}$$

In order to analyze the stability of the overall system, the Lyapunov function V is constructed as

$$V = V_r + \frac{r}{4}\varepsilon_a P\varepsilon_a \tag{42}$$

where $P>0$ is a positive-definite matrix such that $PA_0 + A_0^T P = -I_{n-1}$.

From (41), we obtain

$$\dot{V} \leq \sum_{l=1}^{r} -s_l z_l^2 + (\beta(t)N+1)\dot{\kappa} + \gamma\sigma_s + (r-1)\rho\sigma_p + (u_{\min}+1)\sigma. \tag{43}$$

By integrating the differential inequality (43) on $[0,t]$, we have

$$V(t) \leq V(0) + \int_0^t (\beta(t)N(\kappa)+1)\dot{\kappa}ds - \sum_{l=1}^{r}\int_0^t s_l z_l^2 ds + \Upsilon$$
$$\leq V(0) + \int_0^t (\beta(t)N(\kappa)+1)\dot{\kappa}ds + \Upsilon \tag{44}$$

where $\Upsilon = ((9+m)(r-1)\rho + (7+m)\gamma + u_{\min}+1)\bar{\sigma}$ is a suitable positive constant. By Lemma 2, we know $V(t)$, $\kappa(t)$, and $\int_0^t(\beta N(\kappa)+1)\dot{\tau}ds$ are bounded. According to the definition of $V(t)$, we can obtain that $z_q, q=1,2,\ldots,r$, $\tilde{\gamma}$, $\tilde{\rho}$, and ε_a are bounded, which further implies that y, $\hat{\gamma}$, and $\hat{\rho}$ are bounded. It follows from (8)–(10), it is clear that ξ_0, ξ_1, and ξ_2 are bounded, which leads to that ϱ in (31) is bounded. By following a recursive analysis similar to [27, Ch. 8], according to the boundedness of y, and the smoothness of $\phi_i(y)$, $\psi_i(y)$, and $\Delta(y)$, from Assumption 1, one can get that χ_j, $j=1,\ldots,m$, are bounded. From $\eta_j = A_0^j\chi$, it follows that $\eta_{j,k} = [*,\ldots,*][\chi_1,\ldots,\chi_{j+k}]^T$, which indicates that $\chi_{m+1} = \eta_{m,1} - [*,\ldots,*,1][\chi_1,\ldots,\chi_m]^T$ is bounded. Using the same analytical method as earlier, it can be obtained that χ is bounded, and then, $\hat{x}$ and ε are bounded. Recalling (21)–(25), it can be shown that controller u is bounded. Thus, we prove that all the signals of the closed-loop system are globally uniformly bounded.

From (44), it is verified that $\int_0^t s_1 z_1^2 ds$ is bounded for all $t\geq 0$; by applying the Barbalat's lemma, it is further obtained that $\lim_{t\to\infty} z_1(t) = 0$, which means that y can be regulated asymptotically under deception attacks.

REFERENCES

[1] P. Antsaklis, "Goals and challenges in cyber-physical systems research editorial of the editor in chief," *IEEE Trans. Autom. Control*, vol. 59, no. 12, pp. 3117–3119, Dec. 2014.

[2] Y. Li, L. Shi, P. Cheng, J. Chen, and D. E. Quevedo, "Jamming attacks on remote state estimation in cyber-physical systems: A game-theoretic approach," *IEEE Trans. Autom. Control*, vol. 60, no. 10, pp. 2831–2836, Oct. 2015.

[3] L. An and G.-H. Yang, "Distributed secure state estimation for cyber–physical systems under sensor attacks," *Automatica*, vol. 107, pp. 526–538, 2019.

[4] C. Deng, D. Zhang, and G. Feng, "Resilient practical cooperative output regulation for mass with unknown switching exosystem dynamics under dos attacks," *Automatica*, vol. 139, 2022, Art. no. 110172.

[5] T. Li, Z. Wang, L. Zou, B. Chen, and L. Yu, "A dynamic encryption–decryption scheme for replay attack detection in cyber–physical systems," *Automatica*, vol. 151, 2023, Art. no. 110926.

[6] S. J. Yoo, "Neural-network-based adaptive resilient dynamic surface control against unknown deception attacks of uncertain nonlinear time-delay cyberphysical systems," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 31, no. 10, pp. 4341–4353, Oct. 2020.

[7] W. Chen and Q. Wei, "A new optimal adaptive backstepping control approach for nonlinear systems under deception attacks via reinforcement learning," *J. Autom. Intell.*, vol. 3, no. 1, pp. 34–39, 2024.

[8] X. Jin, W. M. Haddad, and T. Yucelen, "An adaptive control architecture for mitigating sensor and actuator attacks in cyber-physical systems," *IEEE Trans. Autom. Control*, vol. 62, no. 11, pp. 6058–6064, Nov. 2017.

[9] L. An and G.-H. Yang, "Improved adaptive resilient control against sensor and actuator attacks," *Inf. Sci.*, vol. 423, pp. 145–156, 2018.

[10] W. Lin and C. Qian, "Adaptive control of nonlinearly parameterized systems: The smooth feedback case," *IEEE Trans. Autom. Control*, vol. 47, no. 8, pp. 1249–1266, Aug. 2002.

[11] L. Long, Z. Wang, and J. Zhao, "Switched adaptive control of switched nonlinearly parameterized systems with unstable subsystems," *Automatica*, vol. 54, pp. 217–228, 2015.

[12] X.-X. Ren and G.-H. Yang, "Adaptive control for nonlinear cyber-physical systems under false data injection attacks through sensor networks," *Int. J. Robust Nonlinear Control*, vol. 30, no. 1, pp. 65–79, 2020.

[13] J. Zhao and G.-H. Yang, "Fuzzy adaptive finite-time resilient control against unknown false data injection attacks for MIMO nonlinear switched systems with unknown dead zone," *IEEE Trans. Cybern.*, vol. 54, no. 1, pp. 586–598, Jan. 2023.

[14] W. Lv, "Finite-time adaptive neural control for nonlinear systems under state-dependent sensor attacks," *Int. J. Robust Nonlinear Control*, vol. 31, no. 10, pp. 4689–4704, 2021.

[15] J. Zhao and G.-H. Yang, "Fuzzy adaptive secure tracking control against unknown false data injection attacks for uncertain nonlinear systems with input quantization," *Appl. Math. Comput.*, vol. 437, 2023, Art. no. 127551.

[16] X. Wang, Y. Cao, B. Niu, and Y. Song, "A novel bipartite consensus tracking control for multiagent systems under sensor deception attacks," *IEEE Trans. Cybern.*, vol. 53, no. 9, pp. 5984–5993, Sep. 2023.

[17] Z. Liu, J. Tang, Z. Zhao, and S. Zhang, "Adaptive neural network control for nonlinear cyber-physical systems subject to false data injection attacks with prescribed performance," *Philos. Trans. Roy. Soc. A*, vol. 379, no. 2207, 2021, Art. no. 20200372.

[18] M. L. Corradini and G. Orlando, "Robust quantized feedback stabilization of linear systems," *Automatica*, vol. 44, no. 9, pp. 2458–2462, 2008.

[19] L. Xing, C. Wen, Y. Zhu, H. Su, and Z. Liu, "Output feedback control for uncertain nonlinear systems with input quantization," *Automatica*, vol. 65, pp. 191–202, 2016.

[20] X. Yu and Y. Lin, "Adaptive backstepping quantized control for a class of nonlinear systems," *IEEE Trans. Autom. Control*, vol. 62, no. 2, pp. 981–985, Feb. 2017.

[21] L. Zhang and G.-H. Yang, "Observer-based adaptive decentralized fault-tolerant control of nonlinear large-scale systems with sensor and actuator faults," *IEEE Trans. Ind. Electron.*, vol. 66, no. 10, pp. 8019–8029, Oct. 2019.

[22] D. Zhai, L. An, J. Dong, and Q. Zhang, "Output feedback adaptive sensor failure compensation for a class of parametric strict feedback systems," *Automatica*, vol. 97, pp. 48–57, 2018.

[23] J. Zhou, C. Wen, and G. Yang, "Adaptive backstepping stabilization of nonlinear uncertain systems with quantized input signal," *IEEE Trans. Autom. Control*, vol. 59, no. 2, pp. 460–464, Feb. 2014.

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[24] T. Zhang and S. S. Ge, "Adaptive neural network tracking control of MIMO nonlinear systems with unknown dead zones and control directions," *IEEE Trans. Neural Netw.*, vol. 20, no. 3, pp. 483–497, Mar. 2009.
[25] Q. Zhou, P. Shi, J. Lu, and S. Xu, "Adaptive output-feedback fuzzy tracking control for a class of nonlinear systems," *IEEE Trans. Fuzzy Syst.*, vol. 19, no. 5, pp. 972–982, Oct. 2011.
[26] K. K. Hassan, *Nonlinear Systems*. East Lansing, MI, USA: Michigan State Univ., 2002.
[27] M. Krstic, P. V. Kokotovic, and I. Kanellakopoulos, *Nonlinear and Adaptive Control Design*. Hoboken, NJ, USA: Wiley, 1995.
[28] A.-C. Huang and Y.-C. Chen, "Adaptive sliding control for single-link flexible-joint robot with mismatched uncertainties," *IEEE Trans. Control Syst. Technol.*, vol. 12, no. 5, pp. 770–775, Sep. 2004.



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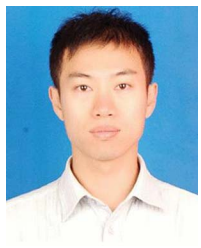
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